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Ovary Cancer Diagnosing Empowered with Machine Learning

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***Abstract*— A high mortality rate is associated with ovarian cancer, one of the most common types of cancers in women. Ovarian cancer refers to a group of disorders that develop in the ovaries and spread to the fallopian tubes and peritoneum. Treatment is most effective when ovarian cancer is discovered in its early stages. Machine learning has recently demonstrated that it is capable of better identifying ovarian cancer and its stages. Most modern research studies on ovarian cancer use a single classification model, leading to poor performance in diagnosis. For the detection of ovarian cancer, the highly sophisticated and efficient machine learning algorithms Support vector machine (SVM) and K-Nearest Neighbor (KNN) are employed in this study. Before diagnosing illness, the suggested approach can optimize and standardize data. Experimental results show that SVM has outperformed KNN in both training and validation performance and achieved an accuracy of 98.1% & 97.16% for training and validation respectively. If used in medical diagnosis systems, the proposed model can significantly improve the accuracy of ovarian cancer detection leading to effective treatment and an increase in patient survival rates.**

Keywords: ovary cancer, machine learning, SVM, KNN

I. INTRODUCTION

One of the most frequent diseases in women is ovarian cancer [1]. In 2018, 295,414 women were diagnosed with ovarian cancer, which claimed the lives of 184,799 people worldwide [2]. Because early-stage tumors are frequently asymptomatic, the majority of ovarian cancer patients have advanced disease at the point of diagnosis, leading to a worse long-term survival [3]. Although ovarian tumors are chemo-sensitive and often demonstrate early success against taxane therapy, five-year survival rates in patients with advanced disease range from 60% to 80% [4]. As a result, much work has gone into finding novel ways to predict disease progression and prognosis in this malignancy. In addition to a pelvic exam or transvaginal ultrasound [5], tumor biomarkers such as carbohydrate antigen 125, carbohydrate antigen 72-4 [6], and human epididymis protein 4 [7] are commonly clinically examined to establish ovarian cancer carrier status. Many studies have been conducted to evaluate the performance of these biomarkers and the indices that use them [8], including a dual marker algorithm using HE4 and carbohydrate antigen 125, Index the risk of malignancy based on imaging characteristics, menopausal status, and carbohydrate antigen 125 to make a distinction patient status from ovarian cancer, and index the risk of malignancy based on imaging characteristics, menopausal status, and carbohydrate antigen 125 to distinguish. Ovarian tumors are common in gynecological clinical practice and are classed as benign, borderline, or cancerous. Among them, ovarian cancer is the leading cause of gynecologic cancer death and the fifth leading cause of cancer death in women in the United States [9]. Given that surgical removal of ovarian tumors is still required for a reliable diagnosis, new testing or diagnostic methods are desirable. For medical diagnostics, artificial intelligence (AI) is regarded as a revolutionary diagnostic approach. It has been investigated in terms of medicine. For starters, AI is not the same as traditional computer programming [10]. Using the incoming data and the specified rules, a prior generic programming method generates outputs. Rather, AI may generate rules based on input and output data. Given the current dataset's input and output data, the AI system may deduce hidden laws and patterns in the data [11]. Furthermore, utilizing the newly found rules and patterns, the AI can predict the output from other input data in the future. Several studies in medicine have demonstrated that AI has good diagnostic accuracy, such as in Computed tomography (CT) scans of the head, skin cancer [12], diabetic retinopathy [13], and patient reports. AI applications in gynecology have already been researched, such as automated Papanicolaou diagnosis and digital colposcopy [14]. Given the extraordinary advancement of computer science over the previous decade, the study of computer-aided diagnosis has advanced.

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II. LITERATURE REVIEW

Ovarian cancer is the fifth most common cause of cancer-related mortality in women. 75% of occurrences occur in postmenopausal individuals, with an annual incidence of 40 per 100,000 in people over the age of 50. Early identification of this disease dramatically improves 5-year survival, increasing it from 3% to 90% [15]. Histopathological assessment is the gold standard for diagnosing and classifying ovarian cancer in histological forms. The interpretation of cell morphology distinguishes the distinct forms of ovarian cancer and directs treatment planning [16], which is best done by ovarian tumor pathologists who are experts in the field. However, inter-observer categorization differences have been recorded. Variations in histological interpretation will result in not only incorrect prognosis and inadequate therapies but also a loss of quality of life [18]. Given the variability and richness of cancer survival prediction, recent studies [19, 20] have demonstrated an integrative analysis of patient genomic data and pathology images, as well as high-throughput omics profiling in conjunction with innovative machine learning algorithms (ML), is highly efficacious in ascertaining cancers. Sun et al. used pathological imaging and genetic data to predict the fate of breast cancer. To bring the disparate knowledge from the two modalities together, a multi-core learning strategy was used. The focus obtained an accuracy of 80.22 % and a precision of 72.73 %. In research, M2DP, a multimodal and multitask feature selection technique for cancer detection. After extracting the properties of the gene expression data and pathology pictures, the M2DP model was used to determine the attributes relevant to the diagnosis. AdaBoost employed the specified features for each patient's diagnosis. The method was tested against breast cancer and a lung cancer benchmark, producing accuracy scores of 0.7253 and 0.7080, respectively. Liu et al. suggested a multimodal deep learning model for predicting breast cancer subtypes. The computer gathered the gene's depths characteristics and imaging methods using a deep learning method and then combined the different information using weighted linear agglomeration. The prediction is 88.07% correct. Recent advancements in convolutional neural networks (CNN) and other deep learning models with comparable capabilities have important implications for medical diagnosis. Deep residual CNN was utilized by Kott and colleagues [21] for the histopathologic detection of prostate cancer. The model exhibited an accuracy of 91.5% in classifying the picture regions as benign or cancerous. Harangi et al. [22] classified dermoscopy pictures using Deep GoogleNet Inception. The accuracy was 0.677. Celik et al. [22] employed histopathologic pictures to detect invasive ductal carcinoma using an in-depth technique. The networks DenseNet-161 and ResNet-50 were employed. The DenseNet-161 model had an f-score of 0.9238 and a precision of 0.9157. Many jobs in the realm of medical applications, on the other hand, are predicated on long-term dependencies [23]. RNN models are among the most successful methods for gaining deep insight into longitudinal data. Short-term and long-term memory (LSTM) is an RNN variant that recognizes short- and long-term dependencies in sequential input. Gao et al. [24] employed LSTMs spaced by time-separated gates to identify lung cancer using actual computed tomography images and simulated data. The method received an f-score of 0.8905. Guo et al. [25] obtained an f-score of 0.572 for a disease inference technique that employed a bidirectional LSTM to identify the sequence of symptoms from a discharge summary. Although deep learning models have demonstrated promise diagnostic performance, constructing a viable deep learning model is not simply a time-consuming and worker undertaking. Recent research has also revealed that the expressive capability of the deep neural network is important and has an impact on deep learning. By demonstrating the universal approximation theorem on the rectified linear unit network defined by width, Lu et al. investigated how the width affects the expressivity of neural networks. The authors demonstrated the superiority of profound channels over shallow ones by demonstrating that in the case of a deep learning model, the total number of neurons required to approximate the natural classes of multivariate polynomials with variable m increases only sequentially with m, but immensely when only one hidden layer is allowed. As a result, optimizing model formation is a pressing issue in the field of deep learning. According to recent research, the model's architecture frequently influences its overall performance and necessitates that it be modified automatically. This research indicated that optimizing hyperparameters is still a significant challenge in the construction of deep learning models, particularly CNN. A strategy based on evolutionary algorithms was presented to build the structure of CNN for autonomous medical image interpretation.

As in previous researches mostly work is done on histopathology medical images and in the end, they don’t get such authentic and accurate results using deep learning networks, LSTM, and RNN. So, the purpose of this study is to propose an ML model by using different machine learning algorithms to overcome the issue of accurate prediction results and achieve high prediction accuracy.

In recent time, various research works have been interrelated for disease predictions by using multiple techniques like transfer learning, deep ensemble learning, Computational Intelligence Approaches [26], Hierarchical Fuzzy Inference System [27], Supervised Machine Learning [28], Deep Extreme Machine Learning [29] and Soft Computing Approaches [30] that have opened some new ideas for the researchers.

III. RESEARCH METHODOLOGY

The rapid advancement of artificial intelligence has made diagnosing in all areas of medical science easy and convenient. Several AI approaches can assist identify diseases, but machine learning algorithms are the most effective in diagnosing diseases early using the patient's medical data. Doctors can use machine learning algorithms to spot ovarian cancer at an early stage. The proposed machine learning-enabled ovarian

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cancer diagnosis methodology (Figure 1) comprises three steps. The suggested model accepts medical patient data as an input and passes it through the pre-processing portion in the first phase. In the pre-processing step, missing values are filled using various algorithmic approaches, and outliers of data are removed, following which the data is normalized using data normalization techniques. Pre-processed data is sent for the training in the second step. The model was trained using KNN and SVM machine learning methods. In the last phase, trained KNN and SVM models were used to make predictions based on the patients' pre-processed data. If the patient is diagnosed to have ovarian cancer, she is directed to a specialist.

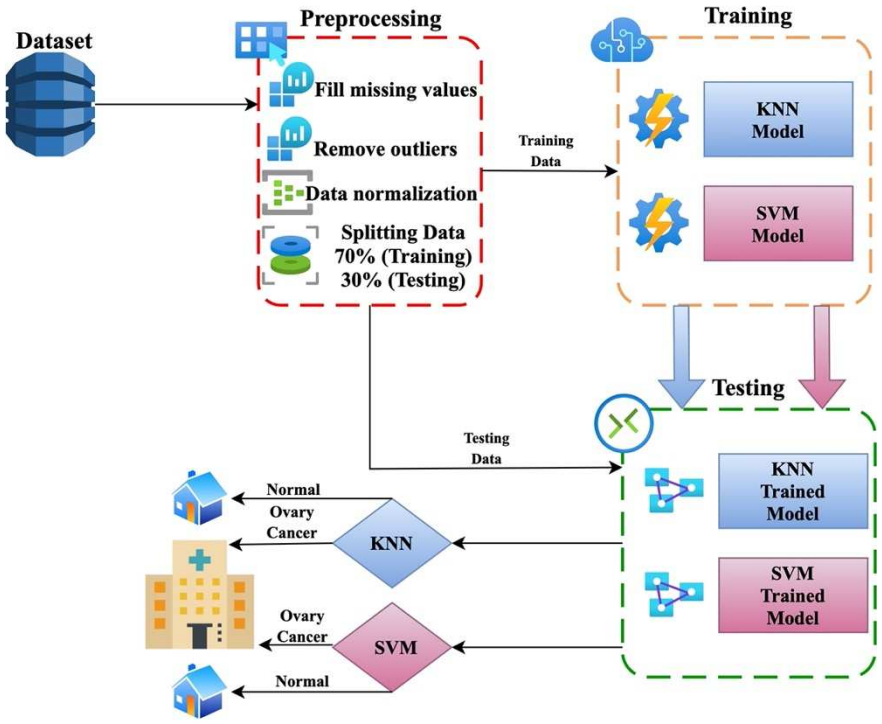


Fig. 1. Proposed model for ovarian cancer diagnosing empowered with machine learning

A. Dataset

The proposed model relied on a dataset for ovarian cancer prediction acquired from Kaggle [34], which is freely accessible. This collection contains data from 463 malignant and non-cancerous patients. The dataset has 48 independent variables and 1 dependent variable and is divided into two classes: ovarian cancer and normal. To prepare data for prediction and training, the suggested model used a variety of missing value approaches, outlier strategies, and normalization techniques.

IV. SIMULATION RESULTS

In this study, MATLAB 2021a is used for the simulation process and the proposed model used KNN and SVM machine learning algorithms for training and prediction of ovarian cancer patients' data. In the pre-processing process, the proposed model divides pre-processed data into 70% for the training process and 30% for the testing process. The proposed model used numerous statistical parameters e-g Classification accuracy (CA), Miss classification rate (CMR), sensitivity, specificity, F1-score, Positive predicted value (PPV), Negative predicted value (NPV), False positive ratio (FPR) to check the performance and authenticity of the proposed model. In the proposed model obtained results ç represents true positive results, € represents true negative results, ∂ represents false-positive results and § represents false-negative results.

$$CA = \frac{ç + €}{ç + € + ∂ + §} * 100 \quad (1)$$

$$MCR = 100 - \frac{ç + €}{ç + € + ∂ + §} * 100 \quad (2)$$

$$Sensitivity = \frac{ç}{ç + §} * 100 \quad (3)$$

$$Specifity = \frac{€}{€ + ∂} * 100 \quad (4)$$

$$F1\text{-}score = \frac{2ç}{2ç + ∂ + §} * 100 \quad (5)$$

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$$PPV = \frac{ç}{ç + \partial} * 100 \quad (6)$$

$$NPV = \frac{€}{€+§} * 100 \quad (7)$$

$$FPR = 100 - \frac{€}{€+\partial} * 100 \quad (8)$$

TABLE I. TRAINING PERFORMANCE OF THE PROPOSED MODEL FOR DIAGNOSING OVARIAN CANCER EMPOWERED WITH MACHINE LEARNING

Instances (322)	True Positive (ç)	True Negative (€)	False Positive (∂)	False Negative (§)
KNN	178	102	21	21
SVM	193	123	6	0

The training performance of the proposed model for diagnosing ovarian cancer empowered with machine learning in table 1 shows that the proposed model used 322 patients' data in the training process and apply KNN and SVM machine learning algorithms to train the models. The KNN algorithm accurately categorizes 280 cases while incorrectly predicting 42 others, and SVM correctly identified 316 patients while incorrectly predicting six others.

TABLE II. TESTING PERFORMANCE OF THE PROPOSED MODEL FOR DIAGNOSING OVARIAN CANCER EMPOWERED WITH MACHINE LEARNING

Instances (141)	True Positive (ç)	True Negative (€)	False Positive (∂)	False Negative (§)
KNN	77	41	16	7
SVM	89	48	4	0

Testing performance of the proposed model for diagnosing ovarian cancer empowered with machine learning is shown in table 2. KNN accurately diagnosed 118 patients and inaccurately identified 23 patients, whereas SVM correctly predicted 137 patients and incorrectly forecasted 4 patients. Consequently, the SVM model is more accurate in diagnosing ovarian cancer and normal patients' outcomes than the KNN model.

TABLE III. TRAINING SIMULATION RESULTS OF PROPOSED MODEL MODEL

Instances (322)	Training (%)	
	KNN	SVM
CA	86.96	98.14
MCR	13.04	1.86
Sensitivity	89.45	100
Specificity	82.93	95.35
F1-Score	89.45	98.47
PPV	89.45	96.98
NPV	82.93	100
FPR	17.07	4.65

Table 3 shows the training results of the proposed model for diagnosing ovarian cancer empowered with machine learning. During training simulation, KNN achieved 86.96%, 13.04%, 89.45%, 82.93%, 89.45%, 89.45%, 82.93%, 17.07% of CA, MCR, sensitivity, specificity, f1-score, PPV, NPV and FPR respectively and SVM achieved 98.14%, 1.86%, 100%, 95.35%, 98.47%, 96.98%, 100%, 4.65% of CA, MCR, sensitivity, specificity, f1-score, PPV, NPV and FPR respectively. It is observed that SVM achieved higher CA and MCR as compared to KNN.

TABLE IV. TESTING SIMULATION RESULTS OF THE PROPOSED

Instances (141)	Testing (%)	
	KNN	SVM
CA	83.69	97.16
MCR	16.31	2.84
Sensitivity	91.67	100
Specificity	71.93	92.31
F1-Score	87.01	97.80
PPV	82.80	95.70
NPV	85.42	100
FPR	28.07	7.69

Table 4 shows the testing results of the proposed model in diagnosing ovarian cancer empowered with machine learning. During testing simulation, KNN achieved 83.69%, 16.31%, 91.67%, 71.93%, 87.01%, 82.80%, 85.42%, 28.07% of CA, MCR, sensitivity, specificity, f1-score, PPV, NPV and FPR respectively and SVM achieved 97.16%, 2.84%, 100%, 92.31%, 97.80%, 95.70%, 100%, 7.69% of CA, MCR, sensitivity, specificity, f1-score, PPV, NPV and FPR respectively. It is observed that SVM achieved superior CA and MCR and performed

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better in other numerous statistical performance parameters in comparison with KNN.

TABLE V. COMPARATIVE ANALYSIS

Model	Training Accuracy (%)	Testing Accuracy (%)
CNN [21]	90.02	88.09
Dense Net [22]	93.4	91.5
Proposed Model	98.14	97.16

Table 5 shows the comparative analysis of the proposed model with previous studies. As mentioned, the proposed model performs better and achieves the highest training and testing accuracy of 98.14% and 97.16% respectively. SVM outclasses the other previously tested algorithms because machine learning algorithms perform far better and fast than most deep learning algorithms.

V. CONCLUSION AND FUTURE WORK

The use of machine learning is becoming increasingly important and constructive in the diagnosis of various diseases using biomarkers, with medical diagnosis improving as a result. This study proposed a machine learning model for the diagnosis of ovarian cancer. The model suggested SVM and KNN methods for the classification of ovarian cancer and normal cases based on the patient's medical history. The proposed methodology includes several statistical metrics for evaluating the model's performance and authenticity. Experimental results show that SVM outperformed KNN attaining 97.16 percent correct diagnostic accuracy and a 2.84 percent miss classification rate. This research will aid in the early and accurate identification of ovarian cancer, allowing patients to get timely and effective treatment. We intend to further expand this research in the future to predict ovarian cancer utilizing additional patient data and various biomarkers using transfer learning and fuzzification.

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